TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders

Standard → slope-intercept: subtract the x-term, then divide every term by the number with y.

Point-slope → slope-intercept: distribute, then add or subtract to get y alone.

Intercepts: x-intercept: put 0 in for y. y-intercept: put 0 in for x.

Solve a formula for a letter: undo everything around that letter, doing the same to both sides.

1. Rewrite the equation in slope-intercept form.

$6x + 3y = 12$

> **Tip:** Get y alone: subtract the x-term, then divide every term.

Subtract 6x: 3y = _____

Divide every term by 3: y = _____

$y = -2x + 4$

$3y = -6x + 12 \rightarrow y = -2x + 4$

2. Rewrite the equation $y - 4 = 3(x + 2)$ in slope-intercept form.

> **Tip:** Distribute first, then add 4 to both sides.

Distribute: y − 4 = _____

Add 4: y = _____

$y = 3x + 10$

$y - 4 = 3x + 6 \rightarrow y = 3x + 10$

3. What are the x- and y-intercepts of $2x - 5y = 10$?

> **Tip:** x-intercept: put 0 in for y. y-intercept: put 0 in for x.

A. (5, 0) and (0, −2)

B. (−5, 0) and (0, 2)

C. (2, 0) and (0, −5)

$y = 0$: $2x = 10$, $x = 5$. $x = 0$: $-5y = 10$, $y = -2$